

Topic Proposal

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ISM I

Topic of Study: Sports Analytics

Date: 9/14/26

I. Introduction and Review of Literature

This research asks: to what extent do MLB-level sabermetric strategies and metrics apply to high school baseball? Sabermetrics has reshaped how professional teams evaluate talent and active players, and that influence is gradually impacting amateur baseball as well. The literature suggest the fit is only partial: Fleisig et al found pitching mechanics stay consistent across levels while velocity output scales sharply with level; Scarborough et al. (2021) found an injury-risk pattern appears in high schoolers almost as often as in professionals; Mair et al. (2022), Matsuura et al (2021), and Kriz et al (2026) each found MLB-style workload rules only work when broken down into components. Caporale and Collier (2014) and McCue et al. (2019) show high school stats alone poorly predict MLB readiness, which Efron and Morris explain statistically as a small-sample problem.

However, no source combines these findings into one usable framework, and that gap is what this research fills. My own team went second in ERA in our district last season and still missed the playoffs, a gap between numbers and outcome I could not let go.

II. Problem Statement & Target Demographics

The problem: high school programs are blindly adopting MLB-derived benchmarks without verifying whether those numbers hold at their level, and the programs least able to verify that gap are the ones adopting it uncritically. The target demographic is under-resourced U.S. high school baseball programs and the players — public teams without booster-funded access to even consumer tools like Rapsodo or HitTrax, whose cost may still be out of most athletic budgets, a widening "data divide" between well-funded programs and public high schools.

The impact: coaches make starting, recruiting, and decisions on unverified MLB intuition (Mair et al., 2022), and players on programs like mine can be strong and still get misread, because the numbers were never built for us.

III. Researcher Qualifications

I bring both statistical training and direct baseball experience to this research. I have completed AP Statistics, AP Calculus AB/BC, AP Physics, along with Sabermetrics 101 and a SABR Sabermetrics Certification, giving me the hypothesis-testing background needed to weigh competing success factors rather than just describe them. I use Claude and Python for modeling on two active projects: umpire strike-call tendency model and a physics-heavy model measuring exit velocity, launch angle, and trajectory. Neither depends on Rapsodo or Trackman-level hardware. As a current varsity player, I also bring firsthand experience of what high school competition and coaching decisions are

organized. Together these let me build a statistically weighted, HS-specific model on accessible tools, test which MLB metrics actually hold up at this level, and keep what a real coach could use.

IV. Conclusion

MLB analytics has transformed how baseball is understood, but there is way less verification work needed to responsibly bring it down to high school baseball, leaving under-resourced programs adopting numbers never built for them. This research does not dismiss analytics as it shows real value to some extent in physics-based modeling and workload design when correctly adjusted, but argues that those adjustments must be done with purpose, factor by factor, without blind assumption. My own team's playoff miss despite having a lead on team ERA is a small example of that large gap. If this research succeeds, it gives coaches and players at programs like mine, without cost-heavy tracking tools, a genuinely usable way to know which numbers to trust and adjust.

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